

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the VBCUA codebase in terms of structure, readability, and reusability. The project follows a modular design so that each major function can be maintained, extended, or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python files follow consistent indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	Modules, data folders, and tests are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as audio_path, similarity_score, and score_result are clear.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like transcribe_audio and calculate_similarity are meaningful.
5	Code Comments	Inline/block comments explain logic	Partial	Core code is understandable; more comments can be added later.
6	Modular Design	Code is split into reusable modules	Yes	Each feature has a separate Python module.
7	No Redundant Code	Duplicate/unused code is removed	Yes	Project is cleaned before GitHub upload.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Audio loading and waveform generation handle invalid files.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	transcription.py	Python / Whisper	Audio-to-text feature	High
2	semantic_analysis.py	Python / Sentence-BERT	Concept similarity evaluation	High
3	audio_analysis.py	Python / Librosa	Audio metrics and waveform	High
4	scoring.py	Python	Final score and feedback generation	High
5	report_generator.py	Python / ReportLab	PDF report creation	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project has a clear folder structure with separate modules.
Readability	4	Function and variable names are meaningful and easy to understand.
Reusability	5	Core logic is reusable across future dashboards or APIs.
Documentation / Comments	4	README and module separation support understanding; more inline comments can be added.
Overall Score	5	The solution is well structured, functional, and ready for demonstration.